DGAT: Dynamic Gaussian Attenuate Transformer for Remote Sensing Image Change Captioning

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***Abstract*—The remote sensing image change captioning (RSICC) technique is designed to enhance geospatial analysis by generating semantic descriptions of differences observed in bi-temporal remote sensing imagery (RSI). Although Transformer-based methods have achieved significant advancements in this field, their standard global attention mechanism allows pixels to pay equal attention to all spatial positions, which lacks explicit prior knowledge of spatial proximity correlation, making the model unable to strengthen local associations and weaken long-distance connections. Furthermore, most existing methods rely on global average pooling when reinforcing channelwise representations, which often dilutes the features of key changed regions by the largely unchanged background, resulting in the inability of feature compression to focus on the critical positions. To address the aforementioned challenges, this article proposes a dynamic Gaussian attenuate Transformer (DGAT), which innovatively introduces a dynamic Gaussian attenuation (DGA) mechanism to model an attenuation law between visual tokens from two perspectives based on Euclidean distance. At the spatial level, DGA constrains visual attention through distance-related Gaussian attenuation, allowing it to prioritize attention on adjacent regions, thereby enhancing the detection of changes in local continuity; at the channel level, DGA identifies the core area based on the visual attention kernel and uses the joint optimization of dynamic Gaussian weighted pooling (DGWP) and channel modulation to focus on features at key positions, effectively enhancing the expression of important channels. The experimental results on three benchmark RSICC datasets demonstrate that our proposed DGAT achieves significantly superior results, verifying the effectiveness of the DGA mechanism. Our code and model weights are available at https://github.com/Pengfei1005/DGAT**

***Index Terms*—Attenuate mechanism, change captioning, channel attention, remote sensing, Transformer.**

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I. Introduction

BENEFITING from the rapid development of deep learning, remote sensing image change detection (RSICD) [1], [2], [3], [4], [5] and classic image captioning (IC) [6], [7], [8], [9], [10] technologies have achieved great success in their respective fields. On this basis, by integrating the spatial positioning ability of RSICD with the semantic expression ability of IC, remote sensing image change captioning (RSICC) [11], [12], [13], [14] has been proposed to generate semantic descriptions of changed regions in dual-temporal remote sensing imagery (RSI), which facilitates a progression from merely detecting changes to comprehensively understanding and explaining them.

In visual tasks, a key prior knowledge is called spatial proximity correlation, which means adjacent pixels in an image usually have a stronger correlation than pixels that are far apart [15], [16]. Convolutional neural networks (CNNs) embody this principle by using small-sized kernels to slide over the image [17], [18], while some variants of Transformers implement this knowledge by confining attention to neighboring regions or windowed areas [19], [20]. Over the past few years, Transformer-based RSICC methods [21], [22], [23], [24] have significantly advanced the field by addressing three critical challenges: 1) modeling long-range dependencies in dual-temporal RSI; 2) capturing complex spatial-semantic relationships; and 3) generating linguistically coherent descriptions of changes. However, these methods all ignore the importance of spatial proximity correlation for modeling the RSICC task. As shown in Fig. 1(a), the standard self-attention mechanism used by the visual encoder in the previous methods globally and indiscriminately calculates the correlation between any two pixels in the image, but it cannot directly perceive the spatial distance between these two positions [25], [26], [27], [28]. This mechanism does not take into account the prior knowledge related to the distance between pixels in RSI, which ignores the influence of spatial distance on the correlation of changes, thus resulting in excessive attention to irrelevant areas. On the other hand, existing methods rarely pay attention to information exchange between channels [29], [30], [31], [32]. As illustrated in Fig. 1(b), previous channel attention mechanisms such as SENet [33] typically use global average pooling during the feature compression process, which causes the unique feature signals of the changed regions to be diluted by the large area of unchanged background, resulting in a complete loss of spatial location information. Some studies on other tasks have begun to explore Gaussian-based

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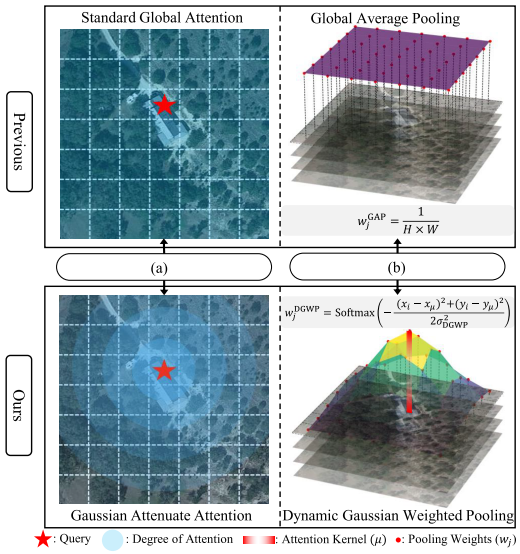

Fig. 1. Comparison of our method with the existing methods in terms of (a) visual attention and (b) pooling operation used for channels.

feature enhancement paradigms or additive position biases to reinforce visual models [34], [35], [36]. However, these methods cannot directly modulate attention between pixels based on their relative spatial distances, nor can they solve the aforementioned limitations regarding channels. Therefore, we propose a dynamic Gaussian attenuate Transformer (DGAT), which applies attenuation to both visual attention and pooling operation to achieve precise modeling on changed features in the RSICC task.

Our proposed DGAT method draws inspiration from Vision RetNet [37], a notable development in natural image processing, which enhances the self-attention mechanism by introducing an explicit spatial prior via a 2-D attenuation based on the Manhattan distance. The DGAT advances the existing RSICC paradigm through two critical innovations. First, we apply a Gaussian attenuation function based on Euclidean distance to directly incorporate the prior knowledge about spatial proximity correlation into the calculation of the visual attention framework. Fig. 1(a) illustrates that our proposed mechanism considers the relative distance between features when evaluating the correlation between them, which enables the model to naturally prioritize changes in local continuous regions and suppress long-distance noise interference. Moreover, we have developed a dynamic Gaussian weighted pooling (DGWP) mechanism driven by the visual attention kernel, which also introduces spatial proximity correlation into the feature compression process. As shown in Fig. 1(b), this pooling operation is no longer an indiscriminate global compression, but focuses on extracting channel information from the most relevant regions according to the kernel with the highest attention level, avoiding the problem that unique feature signals are diluted by background noise. Finally, the channel modulation is combined to realize the effective enhancement of the features at critical positions. The following are the three main contributions of this article.

1) We propose the DGAT method, which is the first to model spatial distance correlation in the RSICC task by introducing a Gaussian attenuation matrix. Our DGAT employs a distance-dependent attenuation rate to allow the model to adaptively compute visual attention and channel weighting for pixels based on their distance in the spatial dimension, thereby precisely localizing changed regions.

2) We develop a dynamic Gaussian attenuation (DGA) mechanism to model the impact of features based on their Euclidean distance. The DGA has two main components to, respectively, enhance the local spatial features and the interaction of channel information, allowing us to map feature-level differences to semantic descriptions in an end-to-end manner.

3) We conduct extensive experiments on three benchmarks to evaluate our DGAT against the state-of-the-art (SOTA) methods, validating its effectiveness in the RSICC task.

The rest of this article is organized as follows. Section II presents some related works. Section III introduces the proposed DGAT method. Subsequently, an analysis of the experimental results is given in Section IV. To conclude, Section V summarizes this research and outlines future works.

II. Related Works

In this section, we introduce previous works on RSICC and positional prior knowledge in the Transformer.

A. Remote Sensing Image Change Captioning

The encoder–decoder framework is the main structure of most RSICC methods. Early works mainly combined a two-stream CNN with recurrent neural networks (RNNs). For example, Hoxha et al. [11] and Chouaf et al. [12] used VGG-16 to extract dual-phase features and generated captions through RNN. However, RNNs have difficulty capturing very long-term dependencies, leading to a relatively low quality of the generated text. With the introduction of Transformer [38], the performance of the RSICC methods has been significantly improved. Liu et al. [13] proposed a dual-branch Transformer with three encoding stages to reinforce the perception of changed areas. They also presented a pure Transformer-structured method [26], which strengthens feature expression by combining scale-aware enhancement modules. To amplify the modeling of the concatenated images, Chang and Ghamisi [14] designed a hierarchical Transformer module, which creatively performed the same attentional operations on independent dual-temporal RSI and the concatenated images. In addition, Zhou et al. [25] chose to extract features directly from the concatenated images and proposed a single-stream extractor to conduct contrastive learning on images. However, these methods suffer from a large number of parameters and high computational complexity. Therefore, Sun et al. [21] developed a lightweight sparse focus Transformer to maintain competitiveness while significantly lowering the number of parameters. Liu et al. [29] explored the application of Mamba with linear complexity in the RSICC task, achieving efficient spatiotemporal joint modeling through time cross scanning. Moreover, Yu et al. [30] was the first to use a diffusion model to decode change captions. Sun et al. [31] applied a frequency-guided filtering module to improve the capacity of the model

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by managing high-frequency noise throughout the diffusion process.

Recently, large model fine-tuning and multitask joint optimization have become prominent research directions. For instance, Liu et al. [28] designed the PromptCC framework, which guides the pretrained large language model (LLM) to generate captions through a multiprompt learning strategy, verifying the advantages of introducing LLMs in the RSICC task. Based on this paradigm, Deng et al. [24] built the first dual-temporal visual-language model to support change captioning generation and change positioning in an interactive question-and-answer manner. Liu et al. [27] proposed a Pix4Cap network that innovatively introduced pseudo-label learning for pixel-level change detection to supervise the RSICC task. However, its performance is marginally compromised due to a data distribution shift. Therefore, they developed the LEVIR-MCI dataset [22] containing both change captioning and change detection data while constructing the MCI model, which used a bi-temporal iterative interaction layer to achieve pixel and semantic alignment. Yang et al. [23] further presented the KCFI framework, which applied a dynamic weighted average strategy to balance the losses between RSICD and RSICC. However, all of the above RSICC methods must learn the correlations of spatial relationships through complex internal nonlinear transformations during training, failing to effectively model the spatial proximity correlation.

B. Positional Prior Knowledge in Transformers

The Transformer architecture processes input sequences in parallel through the self-attention mechanism [38], which computes pairwise similarity between tokens while inherently lacking positional awareness. To deal with this issue, Vaswani et al. [38] provided each word with a unique and fixed positional encoding via periodic trigonometric functions in the original Transformer paper. In contrast, learnable absolute position encodings (LAPEs) [39], [40] initialized a set of trainable embeddings for each position, which were optimized together with word embeddings during the training process to meet the requirements of specific tasks. Furthermore, Shaw et al. [41] proposed relative position encoding to effectively consider the distances between the elements in sequence. On this basis, these works [42], [43] introduced rotational position encoding (RoPE) by incorporating a rotation matrix with positional information into a word vector. However, the drawback of these methods is that they cannot directly perceive the temporal arrangement of the elements. Applying attenuation to elements at different distances is a considered approach. Therefore, Sun et al. [44] proposed RetNet, which enabled language models to implement 1-D temporal attenuation in both training and recursive paradigms. Press et al. [45] developed a linear biased attention to directly apply attenuation weights to different relative distances.

In the field of visual Transformers, the above methods are also applicable. Dosovitskiy et al. [46] assigned LAPE to each image block or patch to indicate its absolute position. To consider the distances between pixels, Wu et al. [47] proposed 2-D relative position encoding with direction awareness. Fan et al. [37] continued to utilize RoPE and extended the 1-D temporal attenuation to two dimensions to introduce rich position priors for visual tasks. Several studies have explored the possibility of introducing spatial or statistical priors by utilizing Gaussian functions. For instance, Luo et al. [34] proposed GAFlow, which used a deformable Gaussian kernel to estimate the optical flow by learning the amplitudes and offsets from data, while Ioannides et al. [35] designed a feature modulation method based on the statistical distribution of input data to fundamentally replace the dot product attention. Ma et al. [36] added a static Gaussian position bias to the attention weight and applied a simplified lightweight self-attention mechanism to execute the hyperspectral image classification task. However, these methods are designed for different tasks and share two shortcomings: they do not directly inject spatial proximity correlation into the Transformer's mechanism, and they cannot prevent irrelevant background information from diluting the key channel features. Therefore, developing novel and adaptive position-aware methods suitable for vision—particularly for RSICC—remains a challenging and nontrivial research problem.

III. Methodology

Considering a dual-temporal RSI (I_1, I_2) as input and the corresponding change caption T as output, the RSICC task can be modeled as the following mapping relationship:

$$T = f_\theta(I_1, I_2) \tag{1}$$

where f is a model parameterized by θ. As shown in Fig. 2, we propose the DGAT method, which consists of three main components: feature extractor, difference encoder, and caption decoder. Initially, the dual-temporal RSI is processed through a Siamese backbone pretrained on ImageNet-1k [48] to extract basic features. These features are then entered into the difference encoder, which uses our proposed DGA mechanism to perform feature enhancement on the changed regions. Finally, the high-level features of the dual-temporal RSI are concatenated and passed into a Transformer decoder to generate change captions.

In Sections III-A–III-C, we will introduce the proposed DGAT method in detail.

A. Feature Extractor

For bi-temporal image processing tasks, it is common to use the structure of a Siamese network with shared weights to extract features [26], [27], [28], [29]. Fig. 2(a) shows that two sub-networks will process different input data in the same way. To maintain a balance between the performance of the model and the efficiency of computation, we adopt the lightweight SegFormer-B1 [49] as the backbone. The feature extraction for two input images I_1 and I_2 can be mathematically defined as

$$X_1, X_2 = \text{Backbone}_{\theta_B}(I_1, I_2) \tag{2}$$

where $X_1, X_2 \in \mathbb{R}^{C\times H\times W}$ represent the bi-temporal extracted features and $\text{Backbone}_{\theta_B}(\cdot)$ denotes the backbone network parameterized by θ_B, which will be fine-tuned during training to better adapt to the RSICC task.

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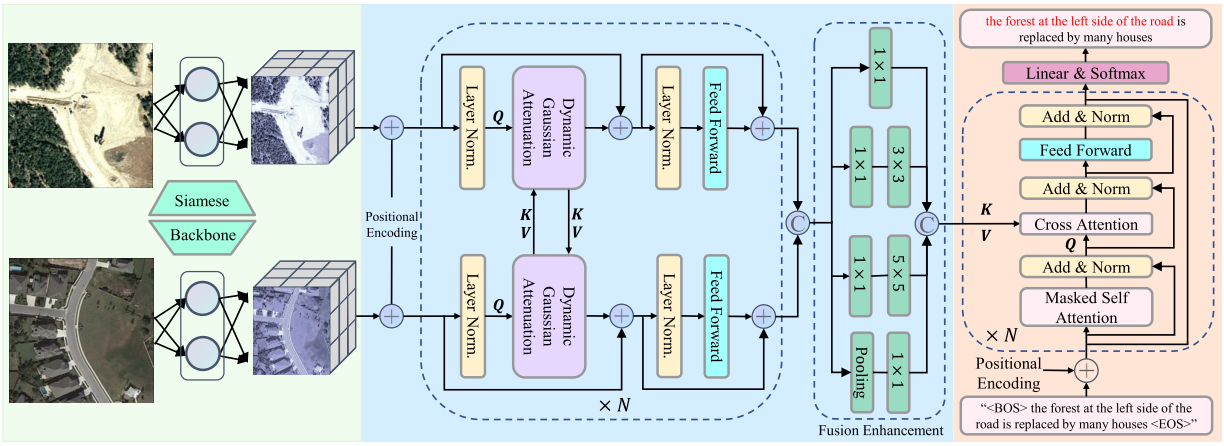

Fig. 2. Overview of the DGAT framework, which contains three main components. (a) Siamese Transformer-based feature extractor. (b) Difference encoder that applies the DGA mechanism. (c) Transformer-based caption decoder.

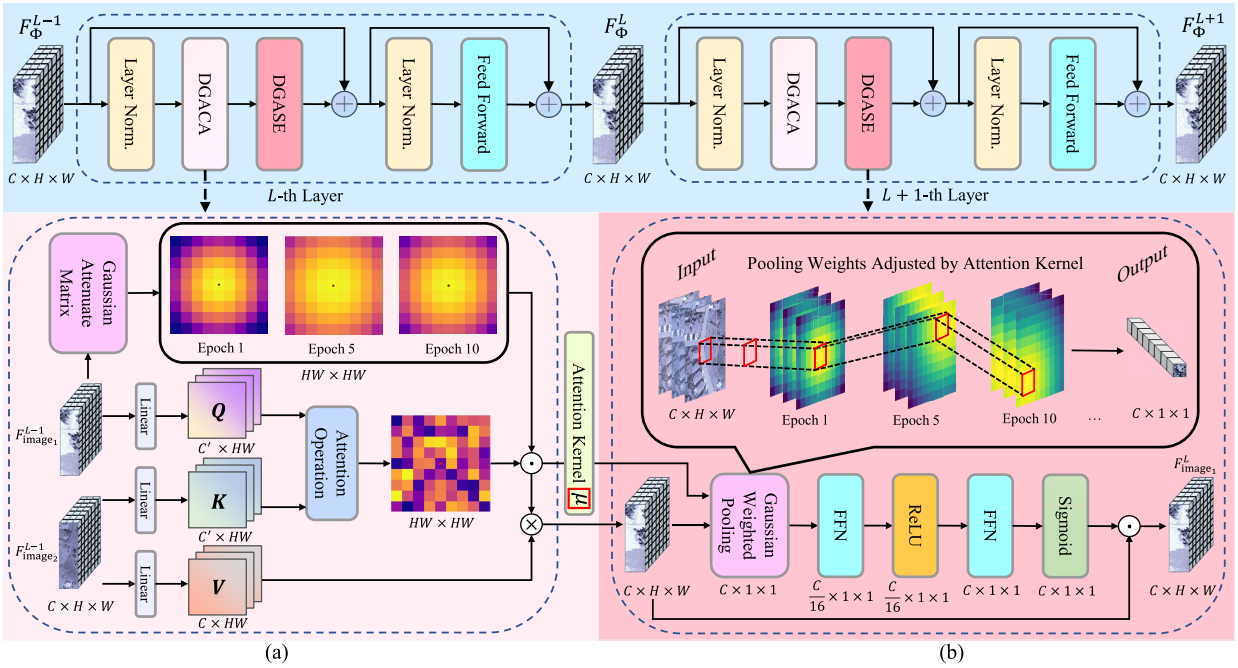

Fig. 3. Overview of our proposed DGA mechanism, which contains two main components: (a) DGACA and (b) DGASE.

B. Difference Encoder

Fig. 2(b) is an overview of the difference encoder, which is the core part of our DGAT method. It consists of a stackable Transformer structure guided by the proposed DGA mechanism, which facilitates the acquisition of feature representations with spatial perception and channel discrimination capabilities from the input images. First, we inject a learnable positional encoding $X_{\text{position}} \in \mathbb{R}^{C\times H\times W}$ into each spatial position of the bi-temporal features to enhance the positional perception ability of the model, i.e., for input features X_1, X_2, we have

$$F_\Phi = X_\Phi + X_{\text{position}}, \quad \Phi = 1, 2. \tag{3}$$

Afterward, the features, F_1 and F_2, are processed by two independent and parallel DGA branches, respectively. Fig. 3 indicates that the DGA mechanism mainly consists of two parts: dynamic Gaussian attenuate cross-attention (DGACA) and dynamic Gaussian attenuate squeeze-and-excitation (DGASE).

1) Dynamic Gaussian Attenuate Cross-Attention: As shown in Fig. 3(a), to explicitly integrate the prior knowledge about "spatial proximity correlation" into the core operation of the model, the DGACA module multiplies the attention score with a Gaussian matrix D^{GA} that attenuates as the Euclidean distance increases, forcing the model to prioritize local information. The ith row of $D^{\text{GA}} \in \mathbb{R}^{HW\times HW}$ is the attenuation relationship of all pixels relative to the ith pixel after the feature map of size $H \times W$ is flattened. Each element D_{ij}^{GA} of D^{GA} in the spatial dimension is calculated as

$$D_{ij}^{\text{GA}} = \exp\left(-\frac{D_{ij}}{2\sigma_{\text{GA}}^2}\right) \tag{4}$$

where

$$D_{ij} = \frac{\|p_i - p_j\|_2^2}{(H-1)^2 + (W-1)^2} \tag{5}$$

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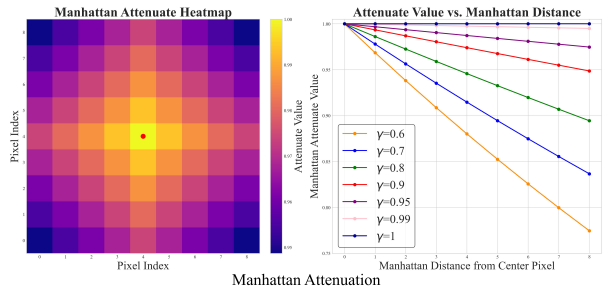

Fig. 4. Manhattan attenuation based on Manhattan distance used by Vision RetNet (taking the center point as an example).

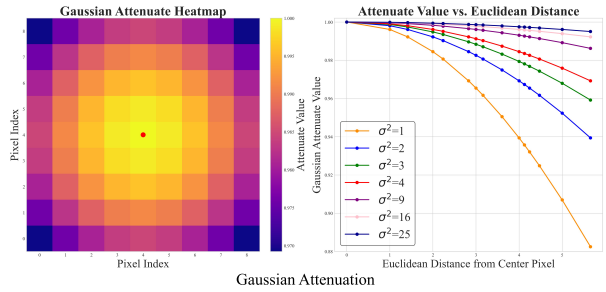

Fig. 5. Gaussian attenuation based on Euclidean distance used by DGAT (taking the center point as an example).

is a normalized square of the pairwise spatial distance between coordinates p_i and p_j located at positions i and $j \in \{1, \ldots, HW\}$, which prevents the attenuation from being too large at long distances. σ^2_{GA} is a DGA factor to control the attenuation degree. It is defined as an adaptive parameter that can be dynamically adjusted through training by the back-propagation algorithm. The range of this parameter is usually restricted, which enables the model to learn the spatial attenuation rate that best suits the current remote sensing dataset.

To incorporate the Gaussian attenuation matrix D^{GA} into the cross-attention framework, we calculate the query (Q_Φ), key (K_Φ), and value (V_Φ) between features F_Φ and $F_{\Phi'}$ of two DGA branches as follows:

$$Q_\Phi = W_{Q_\Phi} F_\Phi, \quad K_\Phi = W_{K_\Phi} F_{\Phi'}, \quad V_\Phi = W_{V_\Phi} F_{\Phi'} \tag{6}$$

where $W_{Q_\Phi} \in \mathbb{R}^{C' \times C}$, $W_{K_\Phi} \in \mathbb{R}^{C' \times C}$, and $W_{V_\Phi} \in \mathbb{R}^{C \times C}$ are three learnable weight matrices. To reduce the computational cost of the attention mechanism, we usually have $C' \ll C$. Ultimately, the DGACA module based on Gaussian attenuate matrix D^{GA} can be given as

$$A_{\mathrm{att}_\Phi} = \mathrm{Softmax}\left(\frac{Q_\Phi^\top K_\Phi}{\sqrt{C'}}\right) \odot D^{\mathrm{GA}} \tag{7}$$

$$F_\Phi^{\mathrm{DGACA}} = V_\Phi A_{\mathrm{att}_\Phi}, \quad \Phi = 1, 2 \tag{8}$$

where $\odot$ represents the Hadamard product, and the Softmax$(\cdot)$ function is used to squash the scores between 0 and 1. A_{att_Φ} is the attenuated attention score used to conduct a weighted summation of the value vector V_Φ to obtain the output F_Φ^{DGACA}.

To demonstrate the advantages of the proposed DGACA, Fig. 4 shows the Manhattan attenuation introduced in Vision RetNet [37], while Fig. 5 reports our novel Gaussian attenuation for comparative analysis. In Manhattan attenuation, each element D_{ij}^{MA} of the attenuated matrix is based on the Manhattan distance between two pixels, defined as

$$D_{ij}^{\mathrm{MA}} = \gamma^{\|p_i - p_j\|_1} \tag{9}$$

where γ is a constant factor. As we can see from (4) and (9), there are two main differences: one is that our Gaussian attenuation uses the square of the Euclidean distance (L_2 distance) while Manhattan attenuation is based on the L_1 distance; another is the DGA factor σ^2_{GA} compared to the constant factor γ. We have found that Gaussian attenuation is superior to Manhattan attenuation in two key aspects: rotation invariance enables accurate modeling of spatial positions, and the gradient characteristic achieves nonlinear attenuation. First, the value of the Manhattan distance is closely related to the orientation of the coordinate axis, and its equidistant lines are squares rotated 45°. The inherent anisotropy of this distance metric leads to a sawtooth effect that introduces an unnatural prior bias to the model, which is related to the image grid. On the contrary, the Euclidean distance used by Gaussian attenuation is isotropic. This rotational invariance allows the model to treat spatial relationships in all directions unbiasedly and fairly, better aligning with the natural distribution of objects in RSI.

In addition, as shown in Fig. 4, the attenuation rate of Manhattan attenuation is proportional to the function value itself: regardless of the position of the current pixel, the weight shrinks at a fixed ratio for each unit increase in distance. This linear attenuation gradient is difficult to adapt to the complex spatial distribution characteristics of the RSI. In contrast, Fig. 5 illustrates that the gradient of our Gaussian attenuation function has the advantage of nonlinearity: in the adjacent area, the attenuation rate is very flat and increases slowly to preserve local details; while in the distant area, the attenuation rate rises rapidly, effectively suppressing background noise. This characteristic can more effectively focus attention on the central area of change.

Through the DGACA module, cross-attention enhances the perception of changes between dual-temporal features. The application of Gaussian attenuation ensures that pixels between nearby positions have a much higher correlation, achieving the purpose of introducing reasonable spatial prior knowledge for attention calculation.

2) Dynamic Gaussian Attenuate Squeeze-and-Excitation: The flowchart of DGASE is shown in Fig. 3(b). In this module, a Gaussian distribution for weighted pooling is dynamically generated through the attention kernel determined by the previous DGACA module, ensuring the model can concentrate on the features of the real changed regions when performing channel compression.

We first aggregate the total attention score for each position $j \in \{1, \ldots, HW\}$ to get a probability distribution d_{j_Φ} as follows:

$$d_{j_\Phi} = \frac{\sum_{i=1}^{HW} A_{\mathrm{att}_\Phi}(i, j)}{\sum_{j'=1}^{HW} \sum_{i=1}^{HW} A_{\mathrm{att}_\Phi}(i, j')} \tag{10}$$

where $\sum_{i=1}^{HW} A_{\mathrm{att}_\Phi}(i, j)$ represents how much attention that each position j receives from all other positions, which reflects the importance of position j in the global scope. This total

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attention score is further normalized to d_{j_Φ} for subsequent weighting operations. Based on the coordinate (x_j, y_j) of each position j, the normalized attention kernel $(x_{\mu_\Phi}, y_{\mu_\Phi})$ is calculated as follows:

$$x_{\mu_\Phi} = \frac{\sum_{j=1}^{HW} d_{j_\Phi} \cdot x_j}{W-1}, \quad y_{\mu_\Phi} = \frac{\sum_{j=1}^{HW} d_{j_\Phi} \cdot y_j}{H-1} \tag{11}$$

where $(x_{\mu_\Phi}, y_{\mu_\Phi})$ means the center coordinate generated based on the attention expectation, representing the position with the highest attention level in the feature map. Next, the pooling weight $w_{j_\Phi}^{\mathrm{DGWP}}$ of each position j in the DGWP is calculated according to the principle that pixels farther away from the kernel have less weight, as follows:

$$w_{j_\Phi}^{\mathrm{DGWP}} = \mathrm{Softmax}\left(-\frac{\left(x_j - x_{\mu_\Phi}\right)^2 + \left(y_j - y_{\mu_\Phi}\right)^2}{2\sigma_{\mathrm{DGWP}}^2}\right) \tag{12}$$

where σ_{DGWP}^2 is another DGA factor adaptively learned via training by the back-propagation algorithm. Afterward, we perform DGWP calculation on the input F_Φ^{DGACA} of (8) by using $w_{j_\Phi}^{\mathrm{DGWP}}$ for each channel in the spatial dimension as follows:

$$F_{\Phi_c}^{\mathrm{DGWP}} = \sum_{j=1}^{HW} F_{\Phi_{j,c}}^{\mathrm{DGACA}} w_{j_\Phi}^{\mathrm{DGWP}} \quad \forall c \in [1, C] \tag{13}$$

$$F_\Phi^{\mathrm{DGWP}} = \left[F_{\Phi_1}^{\mathrm{DGWP}}, F_{\Phi_2}^{\mathrm{DGWP}}, \ldots, F_{\Phi_c}^{\mathrm{DGWP}}\right] \tag{14}$$

where $F_{\Phi_{j,c}}^{\mathrm{DGACA}}$ represents the value of jth position in the space corresponding to the cth channel. DGWP derives a scalar $F_{\Phi_c}^{\mathrm{DGWP}}$ for each channel c by Gaussian weighting all HW pixels in their corresponding space, representing the importance of the current channel. The weight vector $F_\Phi^{\mathrm{DGWP}} \in \mathbb{R}^{C\times 1\times 1}$ is obtained after completing all compression processes. Subsequently, F_Φ^{DGWP} is input into the weight excitation stage of the DGASE module as follows:

$$F_{\mathrm{ex}_\Phi} = \psi\left(W_{\Phi_2} \cdot \delta\left(W_{\Phi_1} \cdot F_\Phi^{\mathrm{DGWP}}\right)\right) \tag{15}$$

$$F_\Phi^{\mathrm{DGA}} = F_\Phi^{\mathrm{DGACA}} \odot F_{\mathrm{ex}_\Phi} \tag{16}$$

where $\delta(\cdot)$ denotes the ReLU activation function, and $\psi(\cdot)$ represents the Sigmoid activation function. Two fully connected layers parameterized by W_{Φ_1} and W_{Φ_2} are used to adaptively reweight each channel to obtain the learned weight $F_{\mathrm{ex}_\Phi} \in \mathbb{R}^{C\times 1\times 1}$. Ultimately, we multiply this weight with the feature F_Φ^{DGACA} channel by channel to generate the final output $F_\Phi^{\mathrm{DGA}} \in \mathbb{R}^{C\times H\times W}$. In this process, the response strength of each channel is dynamically adjusted to highlight key information, as well as suppress redundancy.

DGASE sets the attention kernel generated by the attention expectation as the center, making the weights of all positions follow the Gaussian attenuation as they are farther away from the center, so that the subsequent pooling operation can focus on the key changed regions, thereby highlighting the characteristics of these areas. By combining the spatial sensitivity based on attention distribution with the global dependence of channel dimension, DGASE solves the disadvantage that traditional global average weighting will blur the contribution of essential areas, making a significant improvement in the ability of the overall model to capture critical features.

The entire encoding stage adopts a stackable design, which is mainly composed of a DGA mechanism and a feedforward network (FFN). For the Lth layer, the overall framework can be formulated as

$$Y_\Phi^L = \mathrm{DGA}\left(\mathrm{LN}\left(F_\Phi^L\right)\right) + F_\Phi^L \tag{17}$$

$$F_\Phi^{L+1} = \mathrm{FFN}\left(\mathrm{LN}\left(Y_\Phi^L\right)\right) + Y_\Phi^L \tag{18}$$

where LN$(\cdot)$ represents layer normalization [50], and FFN consists of two linear layers with a ReLU nonlinear transformation. We also use residual connections [51] between each submodule to improve the training stability and expressiveness of the model.

3) Fusion Enhancement Module: The last step of the difference encoder is to fuse the dual-temporal features into one feature to satisfy the interaction requirements of the caption decoder. Several previous works [13], [14], [21], [22] use the same ResBlock in the fusion stage, which is usually composed of multiple serial convolutional layers to perform deep processing of features using a fixed 3×3 convolution kernel. However, the changes in RSI often reveal multiscale characteristics, such as from the appearance of a small house to the development of an entire forest. Therefore, our fusion enhancement module (FEM) adopts a parallel multibranch structure to obtain richer feature representation. The whole process is as follows:

$$F_{\mathrm{fuse}} = \mathrm{Conv}_{1\times 1}\left(\mathrm{Concat}\left(F_1, F_2\right)\right) \tag{19}$$

$$C_1 = \mathrm{Conv}_{1\times 1}\left(F_{\mathrm{fuse}}\right) \tag{20}$$

$$C_2 = \mathrm{Conv}_{3\times 3}\left(\mathrm{Conv}_{1\times 1}\left(F_{\mathrm{fuse}}\right)\right) \tag{21}$$

$$C_3 = \mathrm{Conv}_{5\times 5}\left(\mathrm{Conv}_{1\times 1}\left(F_{\mathrm{fuse}}\right)\right) \tag{22}$$

$$C_4 = \mathrm{Conv}_{1\times 1}\left(\mathrm{MaxPool}\left(F_{\mathrm{fuse}}\right)\right) \tag{23}$$

$$F_{\mathrm{img}} = \mathrm{Concat}\left(C_1, C_2, C_3, C_4\right) \tag{24}$$

where Concat$(\cdot)$ is a concatenation operator, Conv$(\cdot)$ and MaxPool$(\cdot)$ are convolution and maxpooling operators, respectively. As shown in this module, we first concatenate F_1 and F_2, and then a 1×1 convolution kernel is applied to reduce the channel from $2C$ to the dimension C_{dec} used by the decoder while generating the fused feature $F_{\mathrm{fuse}} \in \mathbb{R}^{C_{\mathrm{dec}}\times H\times W}$. This feature is then input into a fusion enhancement block consisting of three CNN branches of different scales (1×1, 3×3, and 5×5 convolution kernel sizes) and a maximum pooling branch for subsequent feature reinforcement. After that, the features C_1, C_2, C_3, and C_4 from four branches are concatenated together along the channel dimension to obtain the final image feature $F_{\mathrm{img}} \in \mathbb{R}^{C_{\mathrm{dec}}\times H\times W}$. This module is the only component of our DGAT method with a CNN architecture. Other than this component, the framework of DGAT is constructed entirely on the Transformer architecture.

C. Caption Decoder

In the text decoding stage, we use the Transformer decoder, which is widely applied in NLP, to generate change captions for the concatenated features. Fig. 2(c) is the integrated structure of the caption decoder. For the original text sequence T_{seq} of length L, we map it to a text embedding $F_{\mathrm{text}} \in \mathbb{R}^{C_{\mathrm{dec}}\times L}$ through a vocabulary-sized embedding matrix. Subsequently,

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F_{text} is input into a stackable Transformer decoder, and the multihead self-attention on the input F_{text} is first calculated to capture dependencies within the text

$$Q_{\text{text}} = W_Q^i F_{\text{text}}, \quad K_{\text{text}} = W_K^i F_{\text{text}}, \quad V_{\text{text}} = W_V^i F_{\text{text}} \tag{25}$$

$$\text{head}^i = V_{\text{text}} \left(\text{Softmax} \left(\frac{Q_{\text{text}}^{\top} K_{\text{text}}}{\sqrt{C_{\text{dec}}/h}} \right) \right) \tag{26}$$

$$\begin{aligned} F_{\text{MH-SA}} &= \text{MH-SA}\left(F_{\text{text}}, F_{\text{text}}, F_{\text{text}}\right) \\ &= W_O \cdot \text{Concat}\left(\text{head}^1, \ldots, \text{head}^h\right) \end{aligned} \tag{27}$$

where W_Q^i, W_K^i, and $W_V^i \in \mathbb{R}^{C_{\text{dec}}/h \times C_{\text{dec}}}$ are three learnable matrices to project F_{text} into different spaces, and then self-attention calculation is performed for each head i. $W_O \in \mathbb{R}^{C_{\text{dec}} \times C_{\text{dec}}}$ is applied to enhance the concatenated multihead output when all the calculations are finished, getting the feature $F_{\text{MH-SA}}$.

The second submodule of the Transformer decoder is multihead cross-attention, which is crucial for the interaction between the concatenated feature of dual-temporal RSI and their corresponding textual feature

$$F_{\text{MH-CA}} = \text{MH-CA}\left(F_{\text{MH-SA}}, F_{\text{img}}, F_{\text{img}}\right). \tag{28}$$

In this stage, the image features F_{img} are utilized to generate key and value matrices to align textual and visual representations for effective cross-modal interaction.

The last submodule is an FFN, which is also composed of two fully connected layers parameterized by W_1^{dec} and W_2^{dec} with ReLU nonlinear activation functions as follows:

$$\text{FFN}\left(F_{\text{MH-CA}}\right) = W_2^{\text{dec}} \cdot \delta\left(W_1^{\text{dec}} \cdot F_{\text{MH-CA}}\right). \tag{29}$$

By stacking N layers of a Transformer decoder, we can get a probability distribution P that can be used to predict words shown as follows:

$$P = \text{Softmax}\left(W_{\text{word}} \cdot \text{Decoder}\left(F_{\text{text}}, F_{\text{img}}\right)\right). \tag{30}$$

Through the vocabulary-sized parameter matrix W_{word}, we map each word from the dimension C_{dec} to the vocabulary size, and then convert the result into a probability distribution after Softmax normalization. During the training phase, we use the image feature F_{img} and the true labels $\{y_1^*, \ldots, y_{t-1}^*\}$ before time step t to conditionally predict the output y_t at each time step t. The prediction result y_t is then compared with the true labely_t^*, and the cross entropy loss L_{CE} is calculated as shown below

$$L_{\text{CE}} = -\sum_{t=1}^{T} \log p\left(y_t \mid y_1^*, \ldots, y_{t-1}^*, F_{\text{img}}\right). \tag{31}$$

In the inference phase, the argmax function is applied to select the word with the highest probability as the final result.

IV. Experiments

We conduct comprehensive experiments and provide illustrative examples on three benchmark datasets to verify the effectiveness of our proposed DGAT method.

A. Dataset

1) LEVIR-CC Dataset: This dataset [13] is derived from the LEVIR-CD [52], a building change detection benchmark containing 637 dual-temporal satellite image pairs with 1024 × 1024 pixels captured across 20 regions in Texas, USA. For the RSICC task, these images were partitioned into 256 × 256 tiles, yielding 10 077 paired samples—half exhibiting changes and half remaining unchanged. By default, the split comprises 6815 training, 1333 validation, and 1929 test pairs. Each pair is annotated with five natural-language captions describing the observed changes, with an average length of 7.99 words and a maximum of 39 words.

2) Dubai-CC Dataset: This dataset [11] captures urban development between 2000 and 2010 via a single dual-temporal satellite image of size 2569 × 2468 pixels. For the RSICC task, 500 paired patches of 50 × 50 pixels were extracted. By default, these are split into 300 training, 50 validation, and 150 test samples. Each patch pair is annotated with five distinct natural-language descriptions, with an average length of 7.35 words and a maximum of 23 words.

3) WHU-CDC Dataset: This dataset [53] is a large-scale dataset built for RSICC and RSICD tasks. It contains 7434 pairs of high-resolution (0.075 m) satellite images with a size of 256 × 256 pixels, which are segmented from image pairs of 32507 × 15354 pixels. By default, these are split into 5947 training, 743 validation, and 744 test samples. Each sample is explained by five sentences with lengths ranging from 3 to 24 and a vocabulary size of 327.

B. Evaluation Metrics

We use BLEU-N [54], METEOR [55], ROUGE-L [56], and CIDEr-D [57] to comprehensively verify the performance of our DGAT method in describing dual-temporal image changes.

1) BLEU-N (N = 1, 2, 3, 4) compares the similarity between the reference caption and the generated caption by counting the proportion of n-grams shared by the two sentences.
2) METEOR introduces semantic matching, such as synonyms or part-of-speech conversion. It will match the generated sentence and the reference sentence word by word based on an external corpus.
3) ROUGE-L uses the $F1$ score to make a balance between recall and precision. This algorithm not only focuses on the matching of words but also on their order when evaluating the generated sentences. It measures the similarity of two sentences by comparing their longest common subsequence.
4) CIDEr-D metric is designed specifically for the IC task. It uses TF-IDF, a weighted technique in information retrieval and text mining, and calculates the cosine similarity of n-grams between the reference caption and the generated caption to evaluate their matching level.

C. Experimental Details

All DGAT experiments were implemented in PyTorch and run on a single NVIDIA RTX 4090 GPU (24 GB). We employ the Adam optimizer [58] with an initial learning rate of 1 ×

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TABLE I

PERFORMANCE COMPARISON OF DGAT AND OTHER SOTA METHODS ON THE LEVIR-CC DATASET. ALL METRICS ARE REPRESENTED AS PERCENTAGES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
RSICCFomer [13]	84.72	76.27	68.87	62.77	39.61	74.12	134.12
PromptCC [28]	83.66	75.73	69.10	63.54	38.82	73.72	136.44
SEN [25]	85.10	77.05	70.01	64.09	39.59	74.57	136.02
RSCaMa [29]	85.79	77.99	71.04	65.24	39.91	75.24	136.56
SFT [21]	84.56	75.87	68.64	62.87	39.93	74.69	137.05
Chg2Cap [14]	86.14	78.08	70.66	64.39	40.03	75.12	136.61
MCI [22]	86.68	78.75	**71.74**	**65.95**	40.80	75.96	140.29
DGAT	**87.09**	**78.93**	71.35	65.30	**40.98**	**76.41**	**141.27**

10^{-4} and a weight decay of 0.95 for 50 epochs. Batch sizes are set to 32 for LEVIR-CC and WHU-CDC, and 8 for Dubai-CC. Both the difference encoder and the caption decoder use a dimension of 512. After each epoch, we judge whether the model has made progress based on its performance on the validation set.

D. Comparison With SOTA Methods

We conduct a comprehensive comparison between our proposed DGAT method and other SOTA methods on three benchmark datasets. These methods include RSICCFomer, PromptCC, SEN, RSCaMa, SFT, Chg2Cap, and MCI. A brief description of each method is as follows.

1) RSICCFomer [13] proposed the widely used LEVIR-CC dataset and is the first Transformer model for the RSICC task. It uses a two-branch Transformer with three encoding stages to enhance the perception of changed regions.
2) PromptCC [28] decouples the RSICC task into two subproblems: whether to change and how to change, which are processed by image-level classifiers and feature-level encoders, respectively. It uses LLMs for the generation of captions for the first time.
3) SEN [25] is the first model to use a single-stream architecture for the RSICC task. It improves the performance of the model by concatenating dual-temporal features and combining contrastive learning strategies.
4) RSCaMa [29] is the first RSICC model to achieve efficient joint spatiotemporal modeling by introducing the Mamba architecture. It proposes the spatial difference-aware and the temporal-traversing state space model to improve spatial and temporal modeling capabilities, respectively.
5) SFT [21] introduces a sparse attention mechanism, which reduces computational overhead by making each pixel perform attention calculations only with the pixels in its row and column.
6) Chg2Cap [14] uses an attention encoder containing a hierarchical self-attention module to model the dual-temporal features and concatenated features in the same way, greatly improving the generalization ability of the model.
7) MCI [22] consists of two branches: pixel-level change detection and semantic-level change description. It introduces a bi-temporal iterative interaction layer to enhance the feature representation abilities.

The experimental results on the LEVIR-CC dataset in Table I show that DGAT exhibits excellent performance on multiple key metrics, highlighting its advantages in the RSICC task. The highest BLEU-1 (87.09) and BLEU-2 (78.93) scores reflect its ability to generate words that are closer to the reference sequence from the word-level accuracy. In addition, the highest METEOR (40.98), ROUGE-L (76.41), and CIDEr-D (141.27) scores reflect its sentence-level generation accuracy from synonym matching, sentence similarity, and semantic consistency, respectively. Although the BLEU-3 (71.35) and BLEU-4 (65.30) scores of DGAT are not as good as the results of MCI, they are only slightly behind and remain ahead of all other methods. Notably, DGAT employs the same backbone and text decoder as MCI, but achieves substantially better overall results. This improvement demonstrates that our DGA mechanism more effectively captures the complex spatial correlations in dual-temporal RSI and aligns them with textual representations in the multimodal embedding space.

As Table II demonstrates, in the Dubai-CC experiments, DGAT achieves superior generalization and outperforms all previous methods on all metrics. Compared to the strongest baseline MCI, DGAT achieved improvements of 3.64%, 3.79%, 4.21%, 8.56%, 6.61%, 4.24%, and 4.01% across all metrics, respectively. This result confirms that even on low-resolution imagery, DGAT can stably generate high-quality textual descriptions. A higher BLEU delivers more accurate word matching at the n-gram level, while higher other metrics indicate that DGAT can faithfully describe changes in dual-temporal RSI at the sentence level. These findings particularly highlight the unique strengths of DGAT in fine-grained change detection and capturing global semantic relationships.

The WHU-CDC dataset is known for its high resolution, which places higher demands on the model's ability to capture subtle changes. The experimental results on the WHU-CDC dataset in Table III further illustrate the strong performance and generalization capabilities of our DGAT in processing high-resolution RSI. It obtains top performance across five key metrics: BLEU-1 (88.37), BLEU-2 (83.50), BLEU-3 (79.25), BLEU-4 (75.96), and ROUGE-L (83.26), and is the suboptimal among all methods on metrics of METEOR (49.43) and CIDEr-D (162.49). This indicates that DGAT can deeply explore the crucial, fine-grained features of change that can only be clearly revealed at high resolution. The consistent high performance across three different RSICC datasets strongly confirms the robustness of our DGAT method. By naturally controlling the attention size and pooling weights of pixels

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TABLE II

PERFORMANCE COMPARISON OF DGAT AND OTHER SOTA METHODS ON THE DUBAI-CC DATASET. ALL METRICS ARE REPRESENTED AS PERCENTAGES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
RSICCformer [13]	67.92	53.61	41.37	31.28	25.41	51.96	66.54
PromptCC [28]	64.25	52.19	43.21	34.59	25.07	53.14	85.11
SEN [25]	70.95	57.28	45.81	36.25	26.62	55.95	91.77
RSCaMa [29]	71.33	58.61	46.99	36.42	28.72	59.86	94.31
SFT [21]	67.30	55.97	47.00	37.30	26.32	56.38	91.59
Chg2Cap [14]	72.04	60.18	50.84	41.70	28.92	58.66	92.49
MCI [22]	73.24	61.17	51.03	40.19	28.73	60.38	101.77
DGAT	**75.91**	**63.49**	**53.18**	**43.63**	**30.63**	**62.94**	**105.85**

TABLE III

PERFORMANCE COMPARISON OF DGAT AND OTHER SOTA METHODS ON THE WHU-CDC DATASET. ALL METRICS ARE REPRESENTED AS PERCENTAGES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
RSICCformer [13]	79.48	74.43	70.94	68.70	42.76	74.80	138.11
PromptCC [28]	84.55	78.44	74.93	72.24	44.77	78.31	146.39
SEN [25]	85.12	80.28	76.84	74.71	46.81	79.45	148.70
RSCaMa [29]	86.39	81.89	77.09	73.61	47.58	80.83	156.24
SFT [21]	84.40	79.81	75.82	72.92	45.60	78.70	150.96
Chg2Cap [14]	83.14	77.16	72.57	69.81	46.42	79.50	149.81
MCI [22]	87.45	82.21	78.43	75.20	**49.74**	82.43	**163.02**
DGAT	**88.37**	**83.50**	**79.25**	**75.96**	49.43	**83.26**	162.49

TABLE IV

COMPARISON OF COMPUTING RESOURCES USED BY DGAT AND OTHER MODELS. THE UNIT OF MEMORY USAGE IS GIGABYTES (GB). THE UNIT OF INFERENCE TIME IS MILLISECONDS (MS). THE UNIT OF THE PARAMETER IS MILLIONS (M). PARAS. MEANS PARAMETERS

Method	Training Memory Usage	Inference Memory Usage	Inference Time	Core Encoder Paras. (1 layer)
SFT [21]	9.43	0.66	31.02	10.52
Chg2Cap [14]	12.17	3.05	34.48	56.69
MCI [22]	11.94	1.99	43.47	85.37
DGAT	**6.64**	**0.62**	**23.54**	**5.07**

based on Gaussian attenuation, it can effectively capture the key information of the changed regions and transform it into high-quality captions.

To evaluate the computational efficiency of our proposed method, we further compare DGAT with several representative baselines on the same hardware, including SFT [21] (a lightweight RSICC method), Chg2Cap [14] (a representative hierarchical RSICC method), and MCI [22] (a RSICC method with the same backbone and decoder). As shown in Table IV, DGAT demonstrates significant advantages across all evaluation metrics. Compared with the lightest method, SFT, our method requires less memory for training and inference. It has a faster inference speed, and the number of parameters of a single core encoder layer is only about half of SFT (5.07 versus 10.52 M). Compared to MCI, the significant difference in computational cost stems almost entirely from the design of a different encoder. The results illustrate that DGAT is far more efficient than MCI. The training memory usage of DGAT has been reduced by nearly half, and its inference memory usage is reduced by two-thirds. In addition, the inference speed of DGAT has increased by approximately 45%, and the number of parameters of the encoder has decreased by 95%. This series of comparisons strongly demonstrates the superior practicality of DGAT. By introducing an explicit spatial prior, it efficiently captures the key changed features with a relatively simple single-layer structure, thereby avoiding the high computational cost caused by stacking complex modules in other methods to enhance the features.

TABLE V

PERFORMANCE COMPARISON OF USING DIFFERENT σ^2_{GA} AND σ^2_{DGWP} ON THE LEVIR-CC DATASET

σ^2_{init}	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D
2	86.80	**65.74**	40.85	76.19	141.19
4	87.08	65.41	40.66	76.07	140.84
9	86.62	65.33	40.55	75.84	140.68
16	86.55	65.28	40.40	75.62	140.07
25	86.32	65.26	40.38	75.26	139.82
1	**87.09**	65.30	**40.98**	**76.41**	**141.27**

E. Ablation Studies

We conduct comprehensive ablation studies on three datasets to verify the effectiveness of each proposed mechanism and module. To keep the table clear and concise, only the five most important metrics are demonstrated, and the presentation of BLEU-2 and BLEU-3 is omitted in the ablation experiments.

1) Parameters Analysis: In our proposed DGA mechanism, σ^2_{GA} and σ^2_{DGWP} control the spatial association of two key mechanisms, respectively. σ^2_{GA} adjusts the receptive field

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TABLE VI

PERFORMANCE COMPARISON OF USING DIFFERENT σ^2_{GA} AND σ^2_{DGWP} ON THE DUBAI-CC DATASET

σ^2_{init}	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D
2	74.56	43.52	29.94	62.45	104.77
4	75.08	**43.74**	**31.18**	62.27	104.35
9	73.43	41.28	30.79	61.14	101.19
16	72.31	38.47	29.01	60.72	101.37
25	72.69	38.90	28.25	60.18	100.66
1	**75.91**	43.63	30.63	**62.94**	**105.85**

TABLE VII

PERFORMANCE COMPARISON OF USING DIFFERENT σ^2_{GA} AND σ^2_{DGWP} ON THE WHU-CDC DATASET

σ^2_{init}	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D
2	**88.45**	75.62	49.06	83.05	162.14
4	87.69	75.18	**49.48**	83.17	161.53
9	86.57	75.35	49.22	82.51	161.28
16	86.30	74.57	48.74	82.43	159.95
25	86.54	74.65	48.36	82.16	160.33
1	88.37	**75.96**	49.43	**83.26**	**162.49**

TABLE VIII

PERFORMANCE COMPARISON USING DIFFERENT CONFIGURATIONS OF LAYERS ON THE LEVIR-CC DATASET

DE.L	CD.L	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	Total Paras.
1	1	**87.09**	65.30	**40.98**	**76.41**	**141.27**	28.31
1	2	85.52	62.98	40.25	75.45	140.29	34.09
1	3	85.87	63.28	40.06	75.58	139.98	39.87
2	1	86.73	65.19	40.44	75.56	140.63	33.37
2	2	85.43	63.45	40.61	75.44	139.71	39.15
2	3	85.34	63.38	40.17	74.95	138.55	44.93
3	1	86.90	**65.75**	40.48	75.88	139.54	38.43
3	2	85.75	64.21	39.43	75.11	138.83	44.21
3	3	85.42	63.72	39.52	74.76	138.20	49.99

of Gaussian attenuated attention in the DGACA module. σ^2_{DGWP} controls the range of Gaussian weighted pooling in the DGASE module. We use the same initial value for σ^2_{GA} and σ^2_{DGWP} for convenience, denoted as σ^2_{init}. The range of σ^2_{init} is set to 1–25 because when σ^2_{init} is 1, it ensures the model has a strong local spatial prior, preventing the initial field of view from being too wide. On the contrary, when σ^2_{init} is 25, the significance of Gaussian attenuation is greatly weakened. At this point, the model approaches the standard attention, thus violating the purpose of introducing the prior knowledge of spatial proximity correlation. This range provides a reasonable initial balance between local detail and global context. The results in Tables V–VII show that when σ^2_{init} is 1, the model achieves optimal or suboptimal performance on most metrics. It indicates that the Gaussian kernel attenuates rapidly, causing the model to focus more on the relationships between closely adjacent pixels, thereby generating a stronger local spatial prior. Therefore, we choose 1 as the initial value of σ^2_{init} in subsequent experiments, which ensures a reasonable attenuation of the visual attention and the DGWP mechanism to effectively capture the spatial feature distribution.

2) Influence of Layers: Both the difference encoder and the caption decoder in our method use a stackable Transformer structure, so that the number of layers in these parts will directly affect the computational cost and performance of the model. We conducted experiments on these two parts with

TABLE IX

PERFORMANCE COMPARISON USING DIFFERENT CONFIGURATIONS OF LAYERS ON THE DUBAI-CC DATASET

DE.L	CD.L	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	Total Paras.
1	1	75.91	**43.63**	**30.63**	62.94	**105.85**	28.31
1	2	74.30	43.25	30.17	62.53	104.40	34.09
1	3	73.19	42.12	29.71	61.79	103.23	39.87
2	1	74.64	42.95	30.28	**63.17**	104.91	33.37
2	2	72.58	40.56	29.36	61.88	103.28	39.15
2	3	72.13	41.71	28.53	59.69	101.65	44.93
3	1	**76.12**	43.31	30.49	61.39	105.54	38.43
3	2	74.36	42.20	28.81	62.47	103.47	44.21
3	3	72.55	41.18	29.07	60.71	102.34	49.99

TABLE X

PERFORMANCE COMPARISON USING DIFFERENT CONFIGURATIONS OF LAYERS ON THE WHU-CDC DATASET

DE.L	CD.L	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	Total Paras.
1	1	**88.37**	**75.96**	**49.43**	**83.26**	162.49	28.31
1	2	86.69	74.64	48.60	81.89	161.56	34.09
1	3	87.62	75.32	49.16	83.13	161.63	39.87
2	1	88.10	75.64	49.08	82.96	159.44	33.37
2	2	87.33	75.06	48.97	81.57	159.70	39.15
2	3	86.85	74.25	47.69	82.49	160.31	44.93
3	1	87.85	75.40	49.41	83.20	**162.79**	38.43
3	2	87.39	75.13	48.95	82.51	159.90	44.21
3	3	87.63	74.37	48.69	82.41	161.05	49.99

TABLE XI

PERFORMANCE COMPARISON OF DIFFERENT POPULAR BACKBONES ON THE LEVIR-CC DATASET

Backbone	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	Backbone Paras.
ResNet-101	86.18	65.25	40.21	75.46	139.97	42.27
VIT/B-16	86.25	**65.63**	40.57	75.59	139.21	86.39
Segformer-B1	**87.09**	65.30	**40.98**	**76.41**	**141.27**	13.15

TABLE XII

PERFORMANCE COMPARISON OF DIFFERENT POPULAR BACKBONES ON THE DUBAI-CC DATASET

Backbone	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	Backbone Paras.
ResNet-101	74.64	42.09	30.20	**63.30**	102.81	42.27
VIT/B-16	73.19	41.20	**31.09**	61.87	98.32	86.39
Segformer-B1	**75.91**	**43.63**	30.63	62.94	**105.85**	13.15

TABLE XIII

PERFORMANCE COMPARISON OF DIFFERENT POPULAR BACKBONES ON THE WHU-CDC DATASET

Backbone	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	Backbone Paras.
ResNet-101	87.02	74.81	48.65	82.24	160.25	42.27
VIT/B-16	87.92	75.39	**49.60**	82.81	**162.98**	86.39
Segformer-B1	**88.37**	**75.96**	49.43	**83.26**	162.49	13.15

different numbers of layers to explore what configuration of layers can achieve a balance between performance and efficiency. In the experimental tables, DE.L denotes the number of layers of the difference encoder, and CD.L represents the number of layers of the caption decoder. The results in Tables VIII–X on three datasets illustrate that as the number of decoder layers increases, the overall performance of the model exhibits a downward trend, which is consistent with the observations of most methods using a Transformer decoder. When both the encoder and decoder are single-layer, the model proves to be the most effective, achieving excellent performance across multiple metrics. This suggests that a single encoder–decoder layer is sufficient for our DGAT method, as deeper layers may lead to information redundancy.

3) Different Backbones: Backbone is the basic network for feature extraction in visual tasks, which has a great impact on the subsequent image-text alignment of the model. Inspired

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TABLE XIV
PERFORMANCE OF DIFFERENT MODULES ON THE LEVIR-CC DATASET

DGACA	DGASE	FEM	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D
×	×	×	82.49	62.69	38.37	73.29	135.45
✓	×	×	85.44	64.69	40.22	75.15	138.06
✓	✓	×	86.34	64.84	40.52	75.83	140.38
✓	✓	✓	**87.09**	**65.30**	**40.98**	**76.41**	**141.27**

TABLE XV
PERFORMANCE OF DIFFERENT MODULES ON THE DUBAI-CC DATASET

DGACA	DGASE	FEM	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D
×	×	×	69.39	37.28	27.06	57.18	86.30
✓	×	×	72.12	40.35	29.34	58.50	94.53
✓	✓	×	73.58	41.28	28.79	61.47	100.82
✓	✓	✓	**75.91**	**43.63**	**30.63**	**62.94**	**105.85**

TABLE XVI
PERFORMANCE OF DIFFERENT MODULES ON THE WHU-CDC DATASET

DGACA	DGASE	FEM	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D
×	×	×	82.30	65.07	41.66	79.18	145.44
✓	×	×	86.27	70.48	47.16	82.13	154.15
✓	✓	×	87.51	75.10	48.64	83.19	160.68
✓	✓	✓	**88.37**	**75.96**	**49.43**	**83.26**	**162.49**

TABLE XVII
PERFORMANCE COMPARISON USING DIFFERENT ATTENUATION MODES (MANHATTAN AND GAUSSIAN) ON THE LEVIR-CC DATASET. ATTENU. REPRESENTS ATTENUATION

Attenu. Type	Attenu. Factor	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D
Manhattan	$\gamma = 0.90$	86.39	65.28	40.26	75.49	140.36
Manhattan	$\gamma = 0.93$	86.23	64.85	40.27	75.27	139.41
Manhattan	$\gamma = 0.95$	86.34	64.74	40.43	75.81	139.72
Manhattan	$\gamma = 0.99$	86.12	65.07	40.21	75.24	139.88
Gaussian	$\sigma^2 = 1$	**87.09**	**65.30**	**40.98**	**76.41**	**141.27**

TABLE XVIII
PERFORMANCE COMPARISON USING DIFFERENT ATTENUATION MODES (MANHATTAN AND GAUSSIAN) ON THE DUBAI-CC DATASET. ATTENU. REPRESENTS ATTENUATION

Attenu. Type	Attenu. Factor	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D
Manhattan	$\gamma = 0.90$	73.79	42.83	30.31	62.71	102.57
Manhattan	$\gamma = 0.93$	72.45	41.73	29.51	60.56	103.74
Manhattan	$\gamma = 0.95$	74.18	40.32	28.73	61.34	98.53
Manhattan	$\gamma = 0.99$	71.52	38.86	28.09	59.27	100.40
Gaussian	$\sigma^2 = 1$	**75.91**	**43.63**	**30.63**	**62.94**	**105.85**

TABLE XIX
PERFORMANCE COMPARISON USING DIFFERENT ATTENUATION MODES (MANHATTAN AND GAUSSIAN) ON THE WHU-CDC DATASET. ATTENU. REPRESENTS ATTENUATION

Attenu. Type	Attenu. Factor	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D
Manhattan	$\gamma = 0.90$	87.74	75.59	48.91	83.13	**162.83**
Manhattan	$\gamma = 0.93$	87.08	74.67	49.06	82.57	161.34
Manhattan	$\gamma = 0.95$	86.64	74.28	48.27	82.72	160.60
Manhattan	$\gamma = 0.99$	86.35	73.74	47.87	82.03	159.76
Gaussian	$\sigma^2 = 1$	**88.37**	**75.96**	**49.43**	**83.26**	162.49

by MCI [22], our standard model employs the SegFormer-B1 [49] with a lightweight Transformer architecture as the feature extractor. To validate the performance of the DGAT method when using different backbones, we compared two widely used backbones: ResNet-101 [51] and ViT/B-16 [46]. The results of the three backbones and their corresponding parameter quantities are demonstrated in Tables XI–XIII. The channel dimensions extracted by ResNet-101, ViT/B-16, and SegFormer-B1 are 2048, 768, and 512, respectively, which significantly affect the number of parameters of the difference encoder and the entire model. It can be observed that ResNet-101 and ViT/B-16 have more parameters, but they do not achieve superior results overall. In contrast, SegFormer-B1 performs relatively better with the least number of parameters (13.15 M) in these datasets. Its lightweight design accurately captures the local details and semantic information of RSI through layer-by-layer feature fusion.

However, it should be noted that the performance of the ResNet version of DGAT is still better than other ResNet methods, such as SFT [21] and Chg2Cap [14], and the performance of the ViT version of DGAT is also better than other ViT methods, such as PromptCC [28] and RSCaMa [29], especially in terms of metrics that reflect the completeness of semantic information. These comparison results verify the effectiveness of our method and prove that the DGA mechanism significantly improves the sensitivity of the model to local changes, thereby achieving better performance.

4) Each Module: We employ a progressive approach, adding key modules to the network one by one to verify the effectiveness of each part. Both Tables XIV–XVI illustrate that the experimental results on the three datasets present a highly consistent trend. When the model does not contain any new modules, it performs the worst. With the addition of the DGACA module, the performance of the model has been significantly improved. Based on this, the DGASE module is further introduced, resulting in even greater performance enhancement. Finally, when these three parts are worked together, all metrics achieve the best results. This demonstrates that with the addition of each module, the ability of the model to generate change captions can be steadily improved, verifying that each part of our DGAT method can play a significant role in improving the performance of the model. By integrating the Gaussian attenuated attention mechanism with the position-sensitive channel compression module and combining them with the multiscale feature fusion mechanism, the model is able to better perceive changes in complex geographical scenes.

5) Attenuate Modes: We replace the Gaussian attenuation based on Euclidean distance with the exponential attenuation based on the Manhattan distance, and try different attenuation factors γ. The results in Tables XVII–XIX on three datasets have indicated that Gaussian attenuation consistently outperforms Manhattan attenuation in almost all metrics. This demonstrates that the Gaussian kernel calculates weights by the square of the Euclidean distance, which can form a smooth weight distribution centered on the focus of attention and simulate spatial dependencies more naturally. The continuity of the Gaussian kernel also enables it to adapt more flexibly to local and global changed patterns of the RSI. In addition, even with Manhattan attenuation, the model performs better than most of the other SOTA methods, which highlights the advantage of introducing a spatially attenuated attention mechanism and a channel adjustment module with a weighted pooling in the RSICC task.

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Fig. 6. (a)–(f) Examples of comparison of the change caption generated by different methods on the LEVIR-CC dataset. Distinctive words are marked in bold.

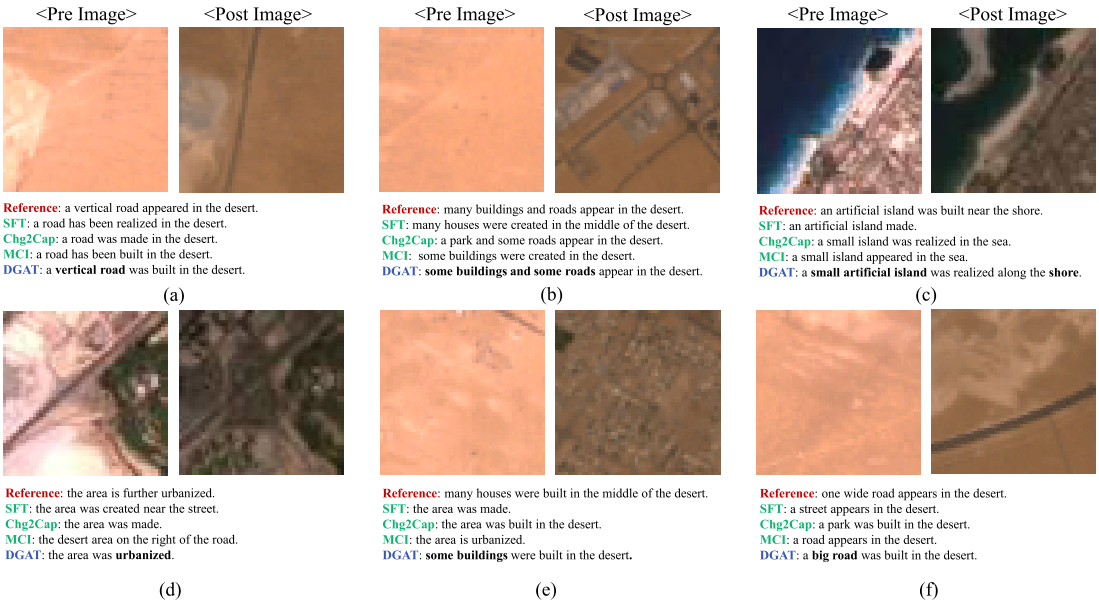

Fig. 7. (a)–(f) Examples of comparison of the change caption generated by different methods on the Dubai-CC dataset. Distinctive words are marked in bold.

F. Qualitative Visualization

To intuitively evaluate our proposed DGAT method, a couple of examples on the LEVIR-CC, Dubai-CC, and WHU-CDC datasets are presented, as shown in Fig. (6)–(8), as well as comparing them with the representative methods in Table IV.

As shown in Fig. 6(a), the proposed DGAT not only describes the appearance of the villa in detail but also accurately identifies that the villa is connected to the road by a path. In contrast, other methods overlook this crucial detail and fail to fully reflect the complexity of scene reconstruction. In Fig. 6(b), DGAT successfully captures that the road and the villas on both sides are newly built, while other methods only capture the newly constructed houses and fail to notice the newly built road. In Fig. 6(c) and (d), DGAT can accurately quantify the location and number of new buildings (e.g., "a building appears in the lower right corner" and "two rows of houses"). On the contrary, other methods either fail to notice the changes or provide ambiguous responses, which highlights DGAT's sensitivity to spatial quantity and positional relationships. Furthermore, in the complex scenes of Fig. 6(e) and (f), DGAT not only captures the rise of the water level through the floor plan but also accurately describes the construction of the road is across the forest, while other methods miss key information. This demonstrates that DGAT can exhibit superior performance compared to other existing methods in perceiving subtle changes.

Although the Dubai-CC dataset presents greater challenges due to its low resolution, our DGAT maintains its superiority. It can be seen that in multiple scenes, DGAT can more accurately identify the types of features and their attributes that have changed. For example, in Fig. 7(a), it accurately captures

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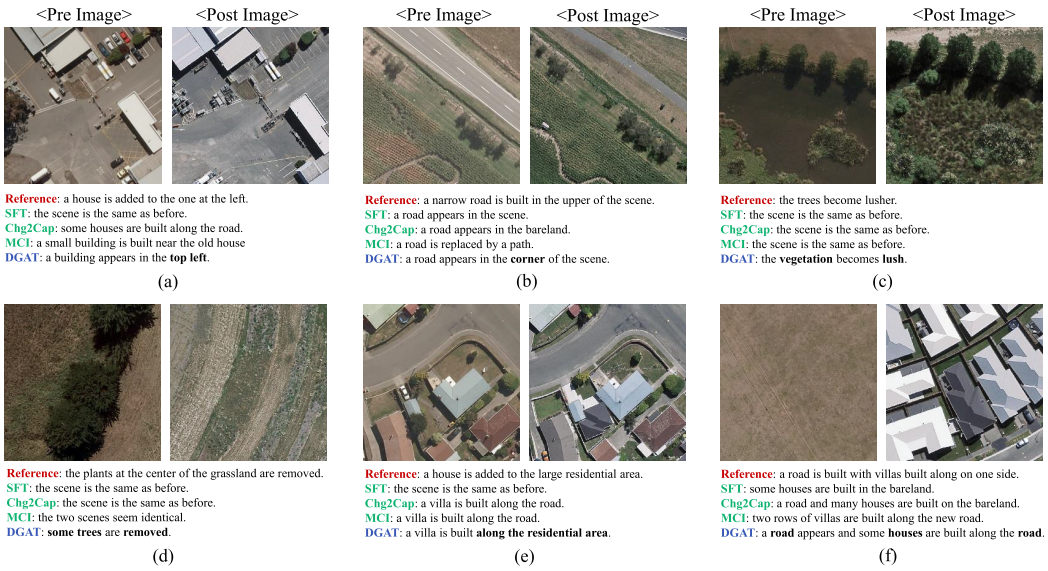

Fig. 8. (a)–(f) Examples of comparison of the change caption generated by different methods on the WHU-CDC dataset. Distinctive words are marked in bold.

the "vertical road," while other methods fail to identify the direction of the road. In Fig. 7(c), it not only identifies the artificial island but also successfully describes its location as "along the shore," but the captions generated by other methods ignore or simplify this critical information. In the remaining examples, DGAT can more vividly describe the changes. In terms of details, it can focus on "a big road" and "some buildings and roads," while it also has the ability to highly abstract these changes into the word "urbanized." In contrast, the sentences generated by other methods deviated slightly from the reference. These examples demonstrate that even for low-resolution RSI, our DGAT method can still effectively focus on the core changed regions in the image, capture more detailed and accurate semantic information, as well as generate high-quality change captions that are consistent with the ground truth.

WHU-CDC provides rich, high-resolution RSI, verifying the ability of DGAT to capture details. In Fig. 8(a), DGAT clearly indicates that the new building is located at the "top left" position; in Fig. 8(b), it locates the newly built road at the "corner." Oppositely, other methods all ignore this crucial information related to the direction, which demonstrates DGAT has a significant advantage in spatial localization accuracy. In Fig. 8(c) and (d), all baseline methods incorrectly judge that the scene has not changed. However, DGAT is the only method that successfully identifies and accurately describes the changes in the status of vegetation (e.g., "lush" and "removed"). This highlights DGAT's superior ability to perceive subtle environmental changes that other methods generally overlook. Furthermore, DGAT also reveals greater comprehensiveness in understanding complex change events. For example, in Fig. 8(e), DGAT successfully describes the presence of a villa along the residential area, which is similar to the reference sentence. In Fig. 8(f), DGAT fully captures the appearance of a road and the construction of new houses along this road, while other methods fail to fully capture both of these aspects simultaneously. These examples certify that DGAT can make full use of the detailed information brought by high-resolution RSI, more accurately capture the subtle changes in the images, and ultimately convert them into corresponding descriptions.

G. Attention Visualization

To further demonstrate the degree of association between words and related image regions during the decoding process using the DGAT method, we visualize the attention weights that the caption decoder focuses on. In Fig. 9, we show the heatmaps of the decoder's attention on the post image when processing a group of samples from the LEVIR-CC, Dubai-CC, and WHU-CDC datasets. When the DGAT is decoding, its attention is observably highly focused and accurately located on the spatial regions in the image that are most relevant to the current generated word (e.g., "road," "villa," "building," and "island"). For high-resolution images with sufficient details, DGAT can accurately focus on specific areas. Even for relatively small changed features in low-resolution images, the method can effectively identify the corresponding areas in a general range. When the generation involves a wider area or background environment (e.g., "houses," "parking," and "desert"), the attention distribution will also expand accordingly to cover a larger range of semantic information. On the other hand, in addition to paying a good attention to nouns, DGAT can also identify the changed regions well when decoding verbs (e.g., "urbanized" and "disappears") related to the changed subject. These visualization results significantly demonstrate that the DGA mechanism brings a powerful visual association ability to the model through Gaussian attenuation. It can not only dynamically adjust the level of attention area according to the target scale but also accurately associate the decoding process with the corresponding changed regions at the semantic level. During the generation of the change captions, it can effectively guide attention to the most relevant

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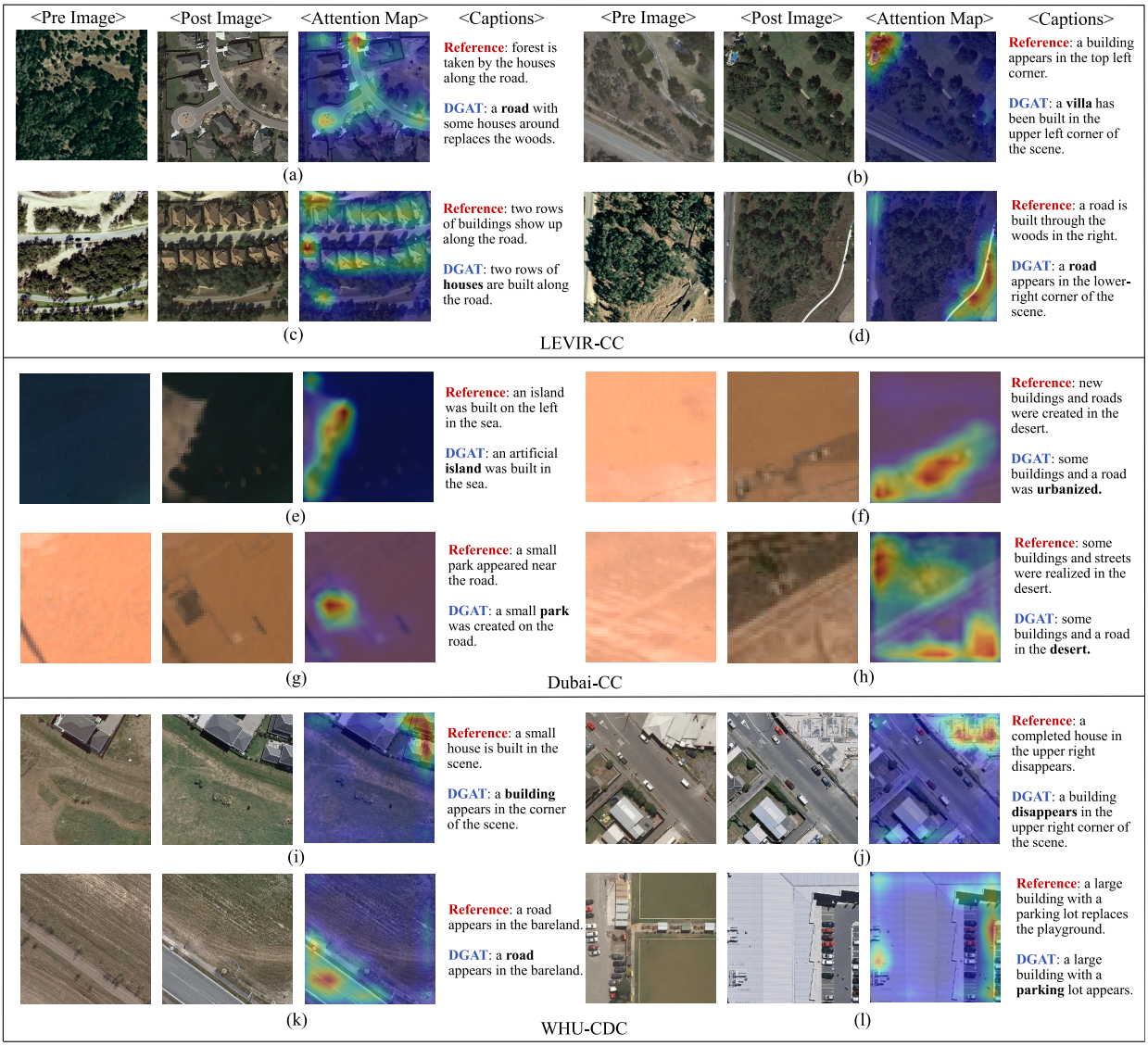

Fig. 9. Examples of visualization of the decoder's attention heatmap shown on the post image from the LEVIR-CC, Dubai-CC, and WHU-CDC dataset. (a)–(d) Examples of the LEVIR-CC dataset. (e)–(h) Examples of the Dubai-CC dataset. (i)–(l) Examples of WHU-CDC dataset. The corresponding words are marked in bold.

area in the image, thereby ensuring the accuracy and relevance of the generated sentences.

V. Conclusion and Future Works

In this study, we propose a novel DGAT method, which effectively solves the key issues of spatial dependence modeling and channel feature refinement in the RSICC task through two main innovative modules. At the spatial level, DGACA directly incorporates the prior knowledge of spatial proximity correlation into the attention mechanism through a 2-D Gaussian attenuation matrix, which explicitly enhances the attention of neighboring regions, enabling the size of visual attention to be adjusted based on distance. At the channel level, DGASE uses DGWP based on the attention kernel to optimize channel feature interactions, improving the ability of the overall model to capture features in key regions. The experiments on the LEVIR-CC, Dubai-CC, and WHU-CDC datasets demonstrate that our proposed DGAT method achieves highly competitive performance in all evaluation metrics and provides stronger support for other remote sensing downstream tasks that require accurate spatial information perception.

Although the proposed DGAT achieves competitive results on the RSICC task, there are still several extensional directions based on its limitations that are worth exploring in the future [59], [60], [61]. First, the current DGAT adopts a CNN-based structure in the fusion module. Exploring a pure and effective Transformer-based algorithm is a worthy research direction, which may help to better capture the long-distance dependencies between features in the fusion stage [59]. Meanwhile, the computational and storage overhead of the attention matrix and Gaussian attenuation matrix in DGAT will increase rapidly as the resolution of RSI becomes higher. Studying more efficient Gaussian attenuation implementations based on windowing or kernel methods is a potential future work [60]. Next, DGAT mainly focuses on the content of changes, while a more intelligent system should also understand how and why things change. Subsequent research may require the use of technologies such as knowledge graphs to associate visual

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changes with reasoning abilities to generate more insightful descriptions [61]. Furthermore, DGAT is currently designed to process dual-temporal homogeneous RSI (e.g., optical-optical). However, the real-world applications often involve multisource and multimodal data. A promising direction is to generalize the DGAT framework to enable it to describe the changes between heterogeneous data pairs (e.g., optical-SAR and optical-LiDAR), thereby providing change analysis for more diverse and complementary RSI [59], [60], [61]. Finally, the most pressing challenge facing the DGAT is that it can only analyze a single change event between two time points, but geographic change in the real world is a continuous process involving many time points. Therefore, future work should explore evolving DGAT from a dual-temporal method to one that can process multiple time series images. This will greatly expand the application scenarios of RSICC technology.

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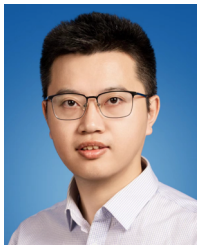

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